

TRULY INNOVATIVE ELECTRONICS

INNOVATION UPDATES

Amongst numerous press releases of new products received by us, these are the ones we found worthy of the title *Truly Innovative Electronics*

Lithium-ion battery leak detection system

IonSniff has launched a lithium-ion battery leak detection system that gives early warnings of potential failures, helping prevent fires in electric



vehicles and electronic devices. The system works by detecting the release

of electrolyte gas in the initial stages of a leak, enabling users to take action before the problem escalates. Once the sensor detects a lithium cell leak, it promptly sounds an alarm, alerting everyone on the premises to follow proper safety procedures and prevent danger. Its proprietary sensors identify the unique gas signature of failing batteries, enabling quick and accurate detection. Extensive lab tests have proven its performance, and it can be installed in homes, garages, or commercial spaces to enhance safety.

IonSniff

<https://www.ionsniff.com>

Single-chip battery fuel gauges

Texas Instruments (TI) has launched the BQ41Z90 and BQ41Z50 single-chip battery fuel gauges for electronics



design engineers, BMS developers, and OEMs building battery-powered products such as laptops, e-bikes, and portable medical equipment.

Featuring adaptive Dynamic Z-Track technology, the gauges deliver state-of-charge and state-of-health accuracy within a 1% error margin, enabling up

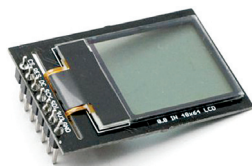
to 30% longer run times than traditional methods. They provide precise, real-time capacity readings even under unpredictable loads, allowing confident battery sizing and reducing oversizing. Dynamic Z-Track continuously self-updates under varying conditions, including AI workloads, to ensure accurate run-time predictions.

Texas Instruments (TI)

<https://www.ti.com>

LCD SPI display module

The 0.8-inch (2.03cm) transparent 48 × 64 LCD SPI display module is available on Tindie and was designed by Questwise Ventures in the United



States. It displays crisp monochrome images on a see-through background,

making it suitable for applications where visuals need to overlay existing views. With a 48 × 64 resolution and black-pixel rendering, it preserves background visibility while presenting clear graphics for both creative and technical uses. The module lacks a built-in backlight, enabling it to be used for optical projection. By shining a bright LED through the screen, users can project sharp silhouettes of the displayed content. The black pixels block light, enabling applications such as miniature projectors and other innovative designs.

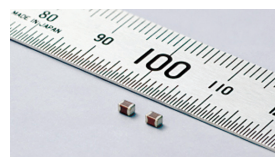
Questwise Ventures in the USA

<https://www.questwiseventures.com>

10μF MLCC

Murata Manufacturing Co., Ltd has begun mass production of its new GCM21BE71H106KE02 multilayer

ceramic capacitor (MLCC), the world's first 0805-inch MLCC to deliver 10μF capacitance with a 50V DC rating, designed specifically for automotive applications. The compact design main-



tains high capacitance, voltage rating, and reliability, represent-

ing a notable step forward in MLCC technology. Optimised for 12V automotive power lines, the capacitor uses Murata's proprietary ceramic materials and thinning technologies to reduce PCB footprint and overall component count. Compared to Murata's previous 0805-inch MLCC, it provides approximately 2.1 times the capacitance in the same size and occupies roughly 53% less space than the older 1206-inch MLCC, offering significant space savings for automotive designs.

Murata Manufacturing Co., Ltd

<https://www.murata.com>

dToF 3D lidar

Orbbec has introduced the Pulsar ME450, a dToF 3D lidar that combines a MEMS mirror and a motor for scanning



patterns and vertical field of view. Supporting multi-pattern scanning in non-repetitive, repetitive non-dense, and dense repetitive modes, it lets robotics systems switch scanning modes on

the fly, removing the need for multiple sensors. The vertical field of view and rotation speeds allow one device to handle obstacle avoidance and mapping, making it suitable for applications such as forklifts, logistics robots, lawn mowers, and surveying equipment.